

Kushal Agrawal

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EDUCATION

Carnegie Mellon University

Master of Science in Computer Engineering

Pittsburgh, PA

Aug 2026 – Dec 2027

University of Wisconsin–Madison

Bachelor of Science in Computer Engineering, GPA: 3.7

Madison, WI

Sep 2022 – May 2026

Coursework: Operating Systems, Data Structures & Algorithms, Distributed Systems, Modern Computer Architecture, Deep Learning, High-Performance Computing, Microprocessor Systems, OOP, Probability, Digital System Design

EXPERIENCE

Google

Madison, WI

Capstone Software Engineer Intern

Feb 2026 – May 2026

- Built IoTriage, a full-stack tool that scans a local network for IoT devices, identifies known vulnerabilities via the NIST NVD, and generates Gemini remediation guidance (Python, Nmap, Express, React).
- Built a rate-limit-aware NVD client with severity and false-positive filtering for accurate vulnerability rankings.

University of Wisconsin–Madison

Madison, WI

Undergraduate Teaching Assistant

Sep 2023 – May 2026

- Led office hours for 850+ students in Digital System Fundamentals, Machine Organization and Intro to Comp. Eng.
- Instructed students in CS 352, CS 354, and CS 252 topics including digital logic design, processor architecture, RTL design, finite state machines, timing analysis, pipelining, C/assembly programming, and memory hierarchy.

Endress+Hauser Group

Greenwood, IN

Embedded Software Engineer Intern

May 2025 – Dec 2025

- Engineered a Dockerized full-stack Netilion IIoT simulator (React/TypeScript, Python/FastAPI) with 30+ REST endpoints, 5 simulation patterns and 200+ health codes, streaming real-time data for hardware-free demos.
- Characterized Bluetooth 6.2 performance on the nRF54L15 via Channel Sounding and multi-device tests, analyzing ranging accuracy, packet reliability, and RF congestion effects for the SGC200 Bluetooth-to-cloud gateway.

Software Engineer Intern

May 2024 – Aug 2024

- Built algorithms to analyze 1,100+ Salesforce dashboards for duplicates, deployed a TypeScript/React naming web tool, integrated it as an extension, and proposed it across 10 departments, potentially saving \$40K annually.
- Designed a custom analog front-end and microcontroller interface to generate and read a 4–20 mA signal from an industrial temperature probe, enabling accurate current-to-temperature conversion for real-time IoT visualization.

PROJECTS

FileHawk.net–Local Semantic Search | *Electron, Python, ChromaDB, SentenceTransformers* | filehawk.net

- Designed and shipped an on-device semantic file retrieval app (Electron/React + Flask) that returns relevant local results in <1s, with all embedding inference running locally—no cloud path for core search.
- Engineered a retrieve-then-rank search pipeline: SentenceTransformers (MS MARCO/All-MiniLM) embeddings queried via ChromaDB, then hybrid ranking signals to score semantic matches across 25+ file types.
- Built incremental indexing (SHA-256 change detection + filesystem watching) that reindexes only changed files.

High-Frequency Currency Arbitrage Engine | *C++, CUDA, OpenMP, Python, CMake* | [GitHub](https://github.com)

- Built a parallel FX arbitrage engine analyzing 86,400 per-second timestamps across 19 currency pairs, modeling markets as dynamic graphs for real-time detection.
- Detected 680 arbitrage opportunities, achieving 0.71% simulated daily ROI with portfolio tracking.
- Leveraged parallelization to cut detection latency by 3× and achieve a 3.5× speedup in arbitrage cycle processing.

FUSE-Based Linux Filesystem | *C, FUSE, RAID 0/1, Python, Bash, GDB* | [GitHub](https://github.com)

- Developed a user-space filesystem in C using FUSE, enabling file creation, deletion, reading, writing, and directory management with inode-based storage and indirect block mapping for large file support.
- Implemented RAID 0 for data striping and RAID 1 for mirroring, ensuring redundancy and fault tolerance.
- Designed and mounted the filesystem with superblock-based metadata tracking and block allocation via bitmaps.

TECHNICAL SKILLS

Languages: C, C++, Python, Java, C#, TypeScript, JavaScript, Bash, SQL, x86 Assembly

Web & Application: React, Node.js, Electron, Next.js, FastAPI, Flask, REST APIs, HTML/CSS, Tailwind CSS

Machine Learning: PyTorch, SentenceTransformers, Hugging Face, ChromaDB, NumPy, Pandas, Matplotlib

Systems & Tools: Linux, Git, Docker, CMake, CUDA, OpenMP, FUSE, GDB, Slurm, AWS, GCP, Postman